The Branch-Order Decomposition of a Backward Readout

An exact identity for the mean branch order, its two-scalar form, and the obstruction to a closed form

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Abstract

A backward readout in a residual network — the gradient of a scalar objective with respect to an internal activation, the object gradient-based attribution methods project into vocabulary space — is a sum over directed paths, each path a product of local Jacobians. The number of such paths is exponential in depth while the cost of computing the readout is linear, so the term count says nothing about how the readout's mass is distributed over paths. *None of what follows is a claim about the function the network computes.* The construction gates each branch with a parameter that leaves the network's forward pass, and hence its output, exactly unchanged; every quantity built from it is therefore a fact about how the objective's gradient is routed among branches, not about how much a branch contributes to that output, and we never use it as a measure of importance. We define the *mean branch order* $\bar{n}$ of a readout, the path length averaged against each order's contribution to the readout itself, and prove four things about it. First, $\bar{n}$ decomposes *exactly* into a sum of per-branch coefficients q_j, with no error term, and non-commutativity of the factors plays no role. Second, each q_j is computable exactly by a single *backward-only ablation* of branch j, so $p := \bar{n}/B = 1 - B^{-1}\sum_j \langle g^{(j\to 0)}, g\rangle / \|g\|^2$ costs one forward and $B+1$ backward passes with no finite-difference step. Third, each q_j admits a closed form in exactly two scalars — a branch-to-skip gain ratio ρ_j and one cosine c_j, both measurable from the same two passes — from which we derive an exact sign law $\operatorname{sign} q_j = \operatorname{sign}(\rho_j + c_j)$, an exact envelope in ρ_j alone, the exact alignment-free equivalence $q_j \lesseqgtr \frac{1}{2} \iff \rho_j \lesseqgtr 1$, and an exact alignment-free upper bound on p. Fourth, the site dependence of q_j is exactly a change of inner product: the difference between two sites is one explicit vector tested against the branch contributions, so it vanishes exactly when that vector is orthogonal to them, and in particular whenever the transport between the sites is conformal. What does *not* follow is a closed form for q_j in intrinsic quantities such as weight norms, layer index or architectural constants; we locate the obstruction precisely and state what would be needed to lift it. On a trained 1.5B-parameter model the substantive consequence of the theorem holds at the single-precision floor, the two-scalar form holds to 1.64×10^{-15} at its worst across 22 inputs, and a median 12.9% of the coefficients are *negative* across inputs, ranging from 4.2% to 24.4%: $q_j \geq 0$ is therefore false, and $\bar{n} \leq B$ holds as a measurement rather than as a theorem. These production-model quantities — the per-layer profile, the coefficient matrix, the negative fraction, the alignment statistics and the sign-law verification — are measured across 22 input prompts, 35,112 coefficients in all, and reported as medians with their full range, because the spread across inputs is large: the branch fraction varies by a factor of nearly three. The sign law records zero violations over all 35,112, and the site-dependence quantiles reported in Section 7 are cross-input as well, so no quantity in this note now rests on a single measurement. The depth ladder below is a separate experiment with three seeds at every depth, and the initialization comparison a third; neither shares these limitations. For arbitrary sequences $\{\rho_j\}, \{c_j\}$, no decay law assumed, we prove a sharp dichotomy when alignment vanishes — from which the pre-norm case falls out as a corollary rather than as the starting point — an absolute-convergence criterion strictly sharper than that envelope, a Dirichlet-type criterion for oscillating alignment via an exact remainder identity, and necessary conditions on the sequence — together with an exact decomposition of the deep-half average of p whose leading term is $\log 2$ and contains no property of the network. Under the idealized pre-norm model of Section 8, the excess sum has an exact closed form via the digamma function, with a derived next-order correction beyond the leading logarithm; fit to the same synthetic sweep, this one-parameter exact model falls short of the two-parameter empirical fit, and we say so rather than reporting only the part that agrees. On a depth ladder trained at fixed width, $L = 4$ to 64, the growth of $\bar{n}$ is logarithmic at initialization and *sub-logarithmic once trained*, and the signed coefficients that make $\bar{n}$ a quasi-mean rather than a mean are created by training rather than by depth. Whether $\bar{n}$ converges we cannot say, and we set out what would settle it.

1 Setting

We work with a residual stack in which each layer contributes one or more *branches* joined additively to a skip path. A pre-norm Transformer block contributes two, in sequence:

$$r_i = x_{i-1} + \mathrm{Attn}_i(x_{i-1}), \qquad x_i = r_i + \mathrm{MLP}_i(r_i),$$

so a stack of L such blocks has $2L$ branches. Nothing below uses the identity of the branch maps; only the additive join structure matters.

Definition 1 (Branch gate)**.** A *branch gate* on a join $y = x + f(x)$ is a parameter $\varepsilon \in \mathbb{R}$ inserted so that the join's *forward* value is unchanged, $y = x + f(x)$ for every ε, while the vector–Jacobian product transporting a cotangent $\bar{y}$ across the join becomes

$$\bar{x} = \bar{y} + \varepsilon\, J_f^{\top}\, \bar{y} = (I + \varepsilon K)\, \bar{y}, \qquad K := J_f^{\top},$$

with J_f the Jacobian of f at the unperturbed activation.

Definition 1 is realizable: an operator that is the identity in the forward pass and multiplies by ε in the backward pass suffices, and interposing one on a branch of an existing graph without disturbing its forward values is an instance of the exact graph surgery of [1]. This is also where the estimand of the whole note is fixed, and it is worth being explicit before a single quantity is defined: since $y = x + f(x)$ for every ε, the network's output does not depend on ε at all, at any branch, for any value. Every quantity built from these gates below is therefore a description of how a gradient is routed through the graph that computes y, and carries no information about y itself beyond the fact of its computation; two networks computing the same function under this construction can have entirely different q_j, and nothing here licenses reading q_j as an amount of credit the branch deserves for the output. Two properties matter and both follow from the forward pass being untouched.

Lemma 2 (Multi-affine exactness)**.** *Fix a read site and let $j = 1, \ldots, B$ index the gated branches lying on some directed path from the read site to the objective. Gate each with its own ε_j. Then the readout at the site is*

$$G(\varepsilon_1, \ldots, \varepsilon_B) \;=\; F_1(\varepsilon_1)\, F_2(\varepsilon_2) \cdots F_B(\varepsilon_B)\, d, \tag{1}$$

where d is the cotangent seeded at the objective and each factor F_j is affine *in ε_j with ε-independent matrix coefficients: $F_j(\varepsilon_j) = I + \varepsilon_j K_j$ for a branch that some contributing path* *bypasses, and* $F_j(\varepsilon_j) = \varepsilon_j C_j$ *for a branch that every contributing path traverses. Consequently* G *is a polynomial map in* $(\varepsilon_1, \ldots, \varepsilon_B)$ *with constant coefficients, of degree at most* 1 *in each variable separately:* G *is* multi-affine.

Proof. Traversing the joins from the objective towards the read site and applying Definition 1 at each gives (1): a gate met on a bypassable join contributes $I + \varepsilon_j K_j$, the identity term being the skip route across that join, while a gate every route must cross contributes no identity term and so has the form $\varepsilon_j C_j$. Because each gate leaves its join's forward value unchanged, every activation in the network — hence every K_j and C_j, hence d — is the one computed by the ungated network and does not depend on any ε_k. Finally ε_j occurs in exactly one factor of (1) and that factor is affine in it, so expanding the product yields a sum of monomials $\prod_{j \in S} \varepsilon_j$ over subsets $S \subseteq \{1, \ldots, B\}$, each with a constant matrix coefficient. $\square$

Multi-affineness, and nothing more, is what every statement in Sections 2–6 uses. In particular the ε_j need not sit in distinct factors: a parallel block, $x_i = x_{i-1} + \mathrm{Attn}_i(x_{i-1}) + \mathrm{MLP}_i(x_{i-1})$, contributes the single factor $I + \varepsilon_A K_A + \varepsilon_M K_M$, which is affine in each of its two variables, and is covered.

Writing $G(\varepsilon \mathbf{1}) = \sum_{k \geq 0} a_k \varepsilon^k$, the coefficient a_k is the sum of the ordered products over all k-element subsets: exactly the total contribution of those backward paths that traverse k *gated branches*. Branch order is not graph-path length — each K_j is itself a sum over many routes inside one branch — so a_k groups graph paths by their branch-traversal count. This grouping is the object of interest.

Definition 3 (Mean branch order)**.** Let $g := G(\mathbf{1})$ be the true readout and assume $g \neq 0$. The *mean branch order* at the site is

$$\bar{n} \;:=\; \frac{\langle g'(1), g \rangle}{\|g\|^2}, \qquad g'(1) := \frac{d}{d\varepsilon}\, G(\varepsilon \mathbf{1})\Big|_{\varepsilon=1}, \tag{2}$$

equivalently $\bar{n} = \frac{1}{2}\, d \log \|G(\varepsilon \mathbf{1})\|^2 / d \log \varepsilon$ at $\varepsilon = 1$. We write $p := \bar{n}/B$ for the *branch fraction.*

Both forms agree: $d/d\log\varepsilon = \varepsilon\, d/d\varepsilon$ equals $d/d\varepsilon$ at $\varepsilon = 1$, and $d \log \|g\|^2 / d\varepsilon = 2\langle g', g \rangle / \|g\|^2$. If a single order k carries all of the readout, then $g(\varepsilon) = \varepsilon^k a_k$ and $\bar{n} = k$ exactly; this anchor is what makes $\bar{n}$ interpretable as an order, and it is used in Remark 11.

2 The exact decomposition

Theorem 4 (Exact per-branch decomposition)**.** *With the notation of Lemma 2, and* $g \neq 0$ *as in Definition 3, set*

$$q_j \;:=\; \frac{\langle \partial_{\varepsilon_j} G|_{\mathbf{1}}, g \rangle}{\|g\|^2}, \qquad \partial_{\varepsilon_j} G|_{\mathbf{1}} \;=\; U_{<j}\, F'_j\, U_{>j}\, d,$$

where F'_j *is the constant matrix* $\partial_{\varepsilon_j} F_j$ *and* $U_{<j} := \prod_{k<j} F_k(1)$, $U_{>j} := \prod_{k>j} F_k(1)$ *are the ordered products of the remaining factors with their gates open. Then*

$$\bar{n} \;=\; \sum_{j=1}^{B} q_j. \tag{3}$$

Proof. By the chain rule for the directional derivative of G along $\mathbf{1}$,

$$g'(1) \;=\; \frac{d}{d\varepsilon} G(\varepsilon\mathbf{1})\Big|_{\varepsilon=1} \;=\; \sum_{j=1}^{B} \partial_{\varepsilon_j} G\big|_{\mathbf{1}}.$$

Differentiating (1) with respect to ε_j replaces the single factor $F_j(\varepsilon_j)$ by F_j' and leaves every other factor untouched, *in place*; the remaining factors are then evaluated at $\varepsilon_k = 1$, giving $U_{<j} F_j' U_{>j} d$. Taking $\langle \cdot, g\rangle / \|g\|^2$ of both sides and using linearity of the inner product in its first argument yields (3). $\square$

Remark 5. Non-commutativity of the factors never enters. This is worth stating because it is the objection one expects: the factors do not commute, and yet the decomposition is exact and order-free. The reason is that ∂_{ε_j} acts on one factor *in situ*, so no reordering is ever performed. Multi-affineness in Lemma 2 is the whole content.

The next statement turns Theorem 4 from a decomposition into a measurement procedure, and is the practical point of this note.

Theorem 6 (Exact ablation identity)**.** *With $g \neq 0$ as in Definition 3, let $g^{(j\to 0)} := G(\mathbf{1} \mid \varepsilon_j = 0)$, the readout with branch j's gate closed and all others open, and put*

$$\beta_j := g - g^{(j\to 0)}, \qquad \sigma_j := g^{(j\to 0)}, \qquad \text{so} \qquad g = \beta_j + \sigma_j. \tag{4}$$

Then, exactly,

$$q_j \;=\; \frac{\langle \beta_j, g\rangle}{\|g\|^2} \;=\; 1 - \frac{\langle g^{(j\to 0)}, g\rangle}{\|g\|^2}, \qquad\qquad p \;=\; 1 - \frac{1}{B}\sum_{j=1}^{B} \frac{\langle g^{(j\to 0)}, g\rangle}{\|g\|^2}. \tag{5}$$

Hence $\bar{n}$ and p are computable from one forward pass and $B{+}1$ backward passes, with no finite-difference step, no extrapolation and no truncation.

Proof. Fix j and regard G as a function of ε_j alone, all other gates set to 1. By Lemma 2 it is affine in ε_j, and the derivative of an affine scalar-to-vector map is its own constant increment, so

$$\partial_{\varepsilon_j} G\big|_{\mathbf{1}} \;=\; G|_{\varepsilon_j=1} - G|_{\varepsilon_j=0} \;=\; g - g^{(j\to 0)} \;=\; \beta_j.$$

Substituting into the definition of q_j in Theorem 4 and using $\langle g, g\rangle/\|g\|^2 = 1$ gives the two forms in (5); summing over j, dividing by B and applying (3) gives the expression for p. $\square$

The decomposition (4) splits the readout at each branch into the part routed *through* that branch and the part routed around it, and it is this splitting, not the ordered product, that the rest of the note uses.

Remark 7 (Why no cheaper exact route exists). The estimator suggested by Definition 3 is a central difference of $\phi(u) := \log \|G(e^u\mathbf{1})\|^2$ at $u = 0$, which carries a truncation error $h^2\phi'''(0)/6 + O(h^4)$; since $G(\varepsilon\mathbf{1})$ has degree B in ε, ϕ is not a low-order polynomial and no fixed two-point scheme evaluates $\phi'(0)$ exactly. Recovering the coefficients a_k instead requires $B{+}1$ evaluations, and doing so from real nodes on the diagonal $\varepsilon_1 = \cdots = \varepsilon_B$ means inverting a Vandermonde matrix of degree B, which for the depths of interest ($B \approx 50$) is hopeless in single precision. Theorem 6 uses the same number of evaluations and is perfectly conditioned, each q_j being read off one inner product. What buys the conditioning is that its evaluation points $\{\mathbf{1}\} \cup \{\mathbf{1} - e_j\}_j$ lie *off* the diagonal: it exploits per-branch access to the gates, which the diagonal scheme forgoes.

3 Two consequences that check an implementation

Corollary 8 (Closed cone). *Extend the definition of q_j in Theorem 4 to every gate in the network. If branch j lies on no directed path from the read site to the objective, then $q_j = 0$ exactly, and $g^{(j\to 0)} = g$.*

Proof. Such a branch does not occur among the factors of (1), so $\partial_{\varepsilon_j} G \equiv 0$. Note where the hypothesis of Lemma 2 is used: closing an upstream gate does not perturb the forward pass, so it leaves every K_k and C_k in (1) unchanged. Were the gate not forward-transparent, an upstream ablation would move the downstream Jacobians and the conclusion would fail. □

Corollary 9 (Mandatory branch). *If every directed path from the read site to the objective traverses branch j_0, then by Lemma 2 the factor F_{j_0} is $\varepsilon_{j_0} C_{j_0}$, so G is* linear *— not merely affine — in ε_{j_0}; consequently $\sigma_{j_0} = g^{(j_0\to 0)} = 0$ and $q_{j_0} = 1$ exactly.*

Proof. With $F_{j_0}(\varepsilon_{j_0}) = \varepsilon_{j_0} C_{j_0}$, setting $\varepsilon_{j_0} = 0$ in the product (1) makes that one factor vanish and hence the whole product vanish, so $g^{(j_0\to 0)} = G(\mathbf{1} \mid \varepsilon_{j_0} = 0) = 0$ directly, and (5) gives $q_{j_0} = 1$. □

For a read site taken to be the output of a branch, captured after its own gate, that branch's gate satisfies the hypothesis of Corollary 9: the site's only route to the objective is through its own join. This fixes the attainable range of branch orders. Let the mL branches of a stack of L blocks with m sequential branches each be numbered in depth order, $\mathrm{pos}(s) = m(i-1) + r$ for the r-th branch of block i. Then

$$B(s) \;=\; mL - \mathrm{pos}(s) + 1 \;=\; m(L-i) + (m - r + 1), \tag{6}$$

since the cone of a branch output is exactly the suffix of that order beginning at the branch itself: the order is total because all branches read and write one residual stream, and the suffix is reached because the stream leaving join $\mathrm{pos}(s)$ enters the input of every later branch. For the pre-norm block, $m = 2$: an attention site at layer i has $r = 1$ and hence $B = 2 + 2(L-i)$, one more than the naive $1 + 2(L-i)$, because it also traverses the gate of the MLP branch of its *own* layer; an MLP site has $r = 2$ and $B = 1 + 2(L-i)$.

Remark 10 (When B is the maximum order). Because the branch order along a single stream is total, every gate in the cone is traversed by some single path, so $B(s)$ is both the cone size and the maximum attainable branch order, and $p = \bar{n}/B$ normalizes by either reading. The two part company when the order is not total: in a parallel block there are still 2 gates per layer, but no single path can traverse both, so the maximum order is $L - i + 1$ while the cone holds $2(L-i)+1$ gates. Statements about p must therefore fix which normalization is meant; we always use the cone size.

Remark 11 (Diagnostic value, honestly weighted). Corollaries 8 and 9 are consequences of the theorem rather than tests added by hand, so a violation indicts an implementation and not a model. They are not of equal strength. Corollary 9 is satisfied by construction if the gate multiplies before the readout is recorded, so it verifies the order of operations inside one gradient rule and the offset in the tabulation of (6) — useful, but weak. Corollary 8 is the substantive one: it asserts that closing an *upstream* gate leaves a downstream readout numerically unchanged, which exercises the path structure itself and, through Lemma 2, the forward-transparency of the gate. Corollary 9 did nevertheless earn its place once: an early measurement reported $\bar{n} = 1.073$

at a site whose attainable order had been recorded as 1, and since $\bar{n} = k$ exactly when a single order k carries the readout, this was a violated identity rather than a curiosity. It forced the derivation (6).

4 A closed form in two scalars, and what follows from it

Theorem 12 (Two-scalar form). *Fix j, assume $g \neq 0$ and $\sigma_j \neq 0$, and let*

$$\rho_j := \frac{\|\beta_j\|}{\|\sigma_j\|}, \qquad c_j := \cos\angle(\beta_j,\ \sigma_j)$$

be the gain ratio *and the* alignment *of branch j at the site. Then*

$$q_j \;=\; \frac{\rho_j\,(\rho_j + c_j)}{1 + 2\rho_j c_j + \rho_j^2}, \qquad \textit{and} \qquad 1 + 2\rho_j c_j + \rho_j^2 = \frac{\|g\|^2}{\|\sigma_j\|^2} > 0. \tag{7}$$

Both scalars are computable from the same two backward passes as q_j.

Proof. By Theorem 6, $\partial_{\varepsilon_j} G|_{\mathbf{1}} = \beta_j$ and $g = \beta_j + \sigma_j$, so

$$q_j = \frac{\langle \beta_j, \sigma_j + \beta_j\rangle}{\|\sigma_j + \beta_j\|^2} = \frac{\langle \beta_j, \sigma_j\rangle + \|\beta_j\|^2}{\|\sigma_j\|^2 + 2\langle \beta_j, \sigma_j\rangle + \|\beta_j\|^2}.$$

Dividing numerator and denominator by $\|\sigma_j\|^2$ and substituting $\langle \beta_j, \sigma_j\rangle = \rho_j c_j \|\sigma_j\|^2$ gives (7); the denominator identity is the same computation read backwards, and it is positive because $g \neq 0$. The two scalars are functions of $\beta_j = g - g^{(j\to 0)}$ and $\sigma_j = g^{(j\to 0)}$ alone. $\square$

In the sequential case of Lemma 2 one has $\beta_j = U_{<j} K_j v_j$ and $\sigma_j = U_{<j} v_j$ with $v_j := U_{>j} d$, so ρ_j is a *transported* branch-to-skip gain ratio and c_j the alignment of the transported branch and skip contributions. The statement above is phrased without $U_{<j}$ and v_j deliberately: it then covers the parallel block, where σ_j collects the skip *and* the sibling branch, and it makes Corollary 9 the boundary case $\sigma_{j_0} = 0$, $\rho_{j_0} = \infty$, $q_{j_0} = 1$.

Four exact consequences follow, none of them requiring an assumption beyond those of Theorem 12 itself: $g \neq 0$ and $\sigma_j \neq 0$, the latter excluding exactly the mandatory branch j_0, whose boundary values $\rho_{j_0} = \infty$, $q_{j_0} = 1$ were recorded above and are not covered by ρ_j, c_j as real numbers. Write $q(\rho, c)$ for the right-hand side of (7).

Corollary 13 (Sign law and monotonicity). $\operatorname{sign} q_j = \operatorname{sign}(\rho_j + c_j)$*; in particular $q_j < 0$ if and only if $c_j < -\rho_j$, which forces $\rho_j < 1$. Moreover $\partial q/\partial c = \rho(1-\rho^2)/(1 + 2\rho c + \rho^2)^2$, so at fixed ρ_j the coefficient increases with alignment when $\rho_j < 1$ and decreases when $\rho_j > 1$.*

Proof. The denominator of (7) equals $\|g\|^2/\|\sigma_j\|^2 > 0$, so the sign is that of $\rho_j(\rho_j + c_j)$ and $\rho_j > 0$. If $c_j < -\rho_j$ then $|c_j| > \rho_j$ and $|c_j| \le 1$ give $\rho_j < 1$. The derivative is $[\rho(1 + 2\rho c + \rho^2) - 2\rho(\rho^2 + \rho c)]/(\cdot)^2 = \rho(1-\rho^2)/(\cdot)^2$. $\square$

Corollary 14 (Exact envelope in the gain ratio alone). *For $\rho_j < 1$, $-\dfrac{\rho_j}{1-\rho_j} \le q_j \le \dfrac{\rho_j}{1+\rho_j}$, and for $\rho_j > 1$, $\dfrac{\rho_j}{1+\rho_j} \le q_j \le \dfrac{\rho_j}{\rho_j - 1}$; each bound is attained exactly at $c_j = \mp 1$. Consequently*

$$q_j < \tfrac{1}{2} \iff \rho_j < 1, \qquad q_j = \tfrac{1}{2} \iff \rho_j = 1, \qquad q_j > \tfrac{1}{2} \iff \rho_j > 1, \tag{8}$$

the middle equivalence holding for every *admissible c_j: at $\rho_j = 1$ the alignment cancels out of (7) entirely.*

Proof. Substituting $u := \rho c \in [-\rho, \rho]$ turns (7) into $(\rho^2 + u)/(1 + \rho^2 + 2u)$, whose derivative in u is $(1-\rho^2)/(1+\rho^2+2u)^2$; it is therefore monotone in u, increasing for $\rho < 1$ and decreasing for $\rho > 1$, so its extrema occur at $u = \pm\rho$, that is $c = \pm 1$, where it takes the values $\rho/(1+\rho)$ and $-\rho/(1-\rho)$. For (8): $\rho/(1+\rho) < \frac{1}{2}$ exactly when $\rho < 1$, which bounds the whole range above by $\frac{1}{2}$ in that case and below by $\frac{1}{2}$ in the other; at $\rho = 1$ direct substitution gives $q = (1+c)/(2+2c) = \frac{1}{2}$ for every $c \neq -1$, and $c = -1$ is excluded since it forces $g = 0$. □

Equation (8) is the sharpest statement available in one scalar: a single measured coefficient decides, with no knowledge of alignment, whether the branch's contribution to the readout exceeds the contribution routed around it. Inverting the envelope gives the corresponding bracket.

Corollary 15 (Exact bracket on the gain ratio)**.** *Let $\tau_j := |q_j|/|1 - q_j|$ for $q_j \neq 1$. Then*

$$q_j < \tfrac{1}{2} \implies \tau_j \le \rho_j < 1, \qquad\qquad q_j > \tfrac{1}{2} \implies 1 < \rho_j \le \tau_j,$$

with equality in either bound exactly when $|c_j| = 1$.

Proof. For $q_j < \frac{1}{2}$ Corollary 14 gives $\rho_j < 1$ and $-\rho/(1-\rho) \le q \le \rho/(1+\rho)$. If $q \ge 0$ the upper bound gives $q(1+\rho) \le \rho$, hence $q \le \rho(1-q)$ and $\rho \ge q/(1-q) = \tau$; if $q < 0$ the lower bound, multiplied by $1 - \rho > 0$, gives $|q|(1-\rho) \le \rho$, hence $|q| \le \rho(1 + |q|)$ and $\rho \ge |q|/(1+|q|) = |q|/(1-q) = \tau$. For $q_j > \frac{1}{2}$, $\rho_j > 1$ and the two bounds give $\rho \le q/(1-q)$ when $q < 1$ and $\rho \le q/(q-1)$ when $q > 1$, both equal to τ. □

Theorem 16 (Exact alignment-free bounds)**.** *Let j_0 be the mandatory branch of a branch-output site with $B \ge 2$ (at $B = 1$ the cone is j_0 alone and $\bar{n} = p \cdot B = 1$ trivially, by Corollary 9, with nothing left to bound). Write $\rho_{\max} := \max_{j \neq j_0} \rho_j$, and let $\bar{q} := (\bar{n} - 1)/(B-1)$ be the mean coefficient over the $B - 1 \ge 1$ bypassable branches. Then:*

(i) if $q_j < \frac{1}{2}$ for every bypassable j — equivalently, by (8), if $\rho_j < 1$ for every bypassable j — then

$$p \;\le\; \frac{1}{B}\left(1 + \sum_{j \neq j_0} \frac{\rho_j}{1+\rho_j}\right) \;\le\; \frac{1}{B}\left(1 + (B-1)\,\frac{\rho_{\max}}{1+\rho_{\max}}\right); \tag{9}$$

(ii) if $\bar{q} < \frac{1}{2}$ then $\rho_{\max} \ge \bar{q}/(1-\bar{q})$, with no hypothesis on the individual branches.

Proof. (i) is Corollary 14 applied termwise together with $q_{j_0} = 1$ from Corollary 9 and (3), the second inequality using monotonicity of $\psi(x) := x/(1+x)$ (distinct from the ϕ of Remark 7). For (ii), split on whether some bypassable $\rho_j \ge 1$. If so then $\rho_{\max} \ge 1 > \bar{q}/(1-\bar{q})$, because $\bar{q} < \frac{1}{2}$ makes the right-hand side less than 1. If not, every $q_j \le \psi(\rho_j) \le \psi(\rho_{\max})$, so $\bar{q} \le \psi(\rho_{\max})$, and applying the increasing inverse $\psi^{-1}(y) = y/(1-y)$ gives the claim. □

The hypothesis of (ii) is a measurement, not an assumption, which is what makes it usable: it converts a measured branch fraction into a rigorous statement about gains. At the layer-1 site of the trained model of Section 7, the median over 22 prompts is $\bar{n} = 6.322$ with $B = 56$, so $\bar{q} = 5.322/55 = 0.0968 < \frac{1}{2}$ and *some bypassable branch in that cone has $\rho_j \ge 0.107$* — with no assumption of isotropy, alignment neglect or homogeneity. The bound is input-dependent because $\bar{n}$ is: across the 22 prompts it ranges from $\rho_{\max} \ge 0.045$ to $\rho_{\max} \ge 0.182$. Note that (i) genuinely needs its hypothesis: a single branch with ρ_j slightly above 1 and c_j near -1 can carry an arbitrarily large q_j, by Corollary 14, and no bound in $\rho_{\max}$ alone survives that.

The envelope also yields a two-sided bound, and with it the only convergence statement we can make without an assumption about alignment.

Theorem 17 (Two-sided bound and a convergence criterion). *Suppose $\rho_j < 1$ for every bypassable branch in the cone. Then $|q_j| \le \rho_j/(1-\rho_j)$, and consequently*

$$|\bar{n} - 1| \;\le\; \sum_{j \ne j_0} |q_j| \;\le\; \sum_{j \ne j_0} \frac{\rho_j}{1-\rho_j}. \tag{10}$$

If the right-hand side of (10) remains bounded as $L \to \infty$, then $\bar{n}$ converges absolutely to a finite limit $\bar{n}_\infty$, and at the deepest site, where $B = 2L$, the branch fraction satisfies $p \sim \bar{n}_\infty/(2L)$. If moreover $\rho_j \le \rho_* < 1$ uniformly, then $\sum_j \rho_j < \infty$ suffices.

Proof. By Corollary 14, q_j lies in $[-\rho_j/(1-\rho_j),\, \rho_j/(1+\rho_j)]$, whose larger endpoint in absolute value is $\rho_j/(1-\rho_j)$; the triangle inequality applied to $\bar{n} - 1 = \sum_{j \ne j_0} q_j$, which uses $q_{j_0} = 1$ from Corollary 9, gives (10). Absolute convergence of a series of reals implies convergence, and $p = \bar{n}/B$ with $B = 2L$ at a layer-one attention site by (6). The uniform case follows from $\rho/(1-\rho) \le \rho/(1-\rho_*)$. □

Two things about Theorem 17 deserve saying. It is *sufficient and not necessary*: the q_j are signed, so $\sum_j q_j$ can converge while $\sum_j |q_j|$ diverges, and in that conditionally convergent case $\bar{n}$ still has a limit but is not robust — its value then rests on cancellation between the positive and negative parts, and a small perturbation of individual coefficients moves it. The quantity a limit claim really needs to control is therefore $\sum_j |q_j|$, or equivalently the cancellation index $\sum_j |q_j| \,/\, |\sum_j q_j|$, and that index is measurable from the same $B{+}1$ passes at no extra cost. Second, the criterion is stated in ρ_j, which is measurable per branch, so whether it holds is a question one settles rather than assumes.

General convergence criteria

Theorem 17 is also not sharp. Its hypothesis $\sum_j \rho_j/(1-\rho_j) < \infty$ is, once $\rho_j \to 0$, equivalent to $\sum_j \rho_j < \infty$: a condition on the *first* power of ρ_j, where the alignment-free case will turn out to need only the second. The following are stated for arbitrary real sequences $\{\rho_j\}$, $\rho_j \ge 0$, and $\{c_j\}$, $|c_j| \le 1$, with no assumed decay law and no relation between them; they follow from the exact form (7) alone.

Proposition 18 (Sharp dichotomy at zero alignment). *If $c_j = 0$ for every j, then $\sum_j q_j < \infty \iff \sum_j \rho_j^2 < \infty$, with no further hypothesis on $\{\rho_j\}$.*

Proof. At $c_j = 0$, (7) reads $q_j = \rho_j^2/(1+\rho_j^2) \ge 0$. Since $1 + \rho_j^2 \ge 1$ we get $q_j \le \rho_j^2$ termwise, so $\sum \rho_j^2 < \infty$ implies $\sum q_j < \infty$ by comparison. Conversely suppose $\sum q_j < \infty$; then $q_j \to 0$, and since $x \mapsto x/(1+x)$ is a strictly increasing bijection of $[0,\infty)$ onto $[0,1)$, $\rho_j^2 \to 0$. Choose J with $\rho_j^2 \le 1$ for $j \ge J$. Rearranging $q_j(1+\rho_j^2) = \rho_j^2$ gives $\rho_j^2 = q_j(1+\rho_j^2) \le 2q_j$ for $j \ge J$, so $\sum_{j \ge J} \rho_j^2 \le 2\sum_{j \ge J} q_j < \infty$; the remaining finitely many terms are finite. □

Proposition 19 (Absolute convergence at general alignment). *If $\sum_j \rho_j^2 < \infty$ and $\sum_j \rho_j |c_j| < \infty$, then $\sum_j |q_j| < \infty$.*

Proof. $\sum \rho_j^2 < \infty$ forces $\rho_j \to 0$, so fix J with $\rho_j \le \frac{1}{4}$ for $j \ge J$. For such j, using $c_j \ge -1$ and $\rho_j^2 \ge 0$, $D_j = 1 + 2\rho_j c_j + \rho_j^2 \ge 1 - 2\rho_j \ge \frac{1}{2}$, whence by (7) $|q_j| = |\rho_j c_j + \rho_j^2|/D_j \le 2\rho_j|c_j| + 2\rho_j^2$. Summing over $j \ge J$ and adding the finite head gives the claim. □

Proposition 19 is strictly weaker in hypothesis than Theorem 17: $\sum \rho_j < \infty$ implies both of its conditions, while $\rho_j = j^{-3/5}$, $c_j = j^{-1/2}$ satisfies both and has $\sum \rho_j = \infty$. It also contains the sufficient half of Proposition 18, since $c_j \equiv 0$ makes $\sum \rho_j |c_j| = 0$. The next statement covers alignments that do not vanish at all, provided they oscillate.

Proposition 20 (Conditional convergence under oscillating alignment)**.** *Suppose $\rho_j \downarrow 0$ monotonically, $\sum_j \rho_j^2 < \infty$, and the partial sums $C_N := \sum_{j\le N} c_j$ are bounded. Then $\sum_j q_j$ converges.*

Proof. From (7), exactly,

$$q_j - \rho_j c_j - \rho_j^2 \;=\; (\rho_j c_j + \rho_j^2)\Big(\tfrac{1}{D_j} - 1\Big) \;=\; -\,\frac{(\rho_j c_j + \rho_j^2)(2\rho_j c_j + \rho_j^2)}{D_j} \;=:\; r_j.$$

Fix J with $\rho_j \le \frac{1}{4}$ for $j \ge J$, so $D_j \ge \frac{1}{2}$ as above, $|\rho_j c_j + \rho_j^2| \le \frac{5}{4}\rho_j$ and $|2\rho_j c_j + \rho_j^2| \le \frac{9}{4}\rho_j$; hence $|r_j| \le \frac{45}{8}\rho_j^2$ for $j \ge J$, and $\sum_j |r_j| < \infty$ by hypothesis. The series $\sum_j \rho_j c_j$ converges by Dirichlet's test, its two hypotheses being exactly $\rho_j \downarrow 0$ and C_N bounded. Then $\sum_j q_j = \sum_j \rho_j c_j + \sum_j \rho_j^2 + \sum_j r_j$ converges as a sum of three convergent series. □

Proposition 21 (Necessary conditions)**.** (i) *If* $\limsup_j \rho_j < 1$ *and* $\sum_j q_j$ *converges, then* $\rho_j(\rho_j + c_j) \to 0$. *In particular* $\rho_j \to 0$ unless $c_j + \rho_j \to 0$ *along a subsequence on which* ρ_j *stays bounded away from* 0. (ii) *If $q_j \ge 0$ for all large j, q_j is eventually non-increasing, and $\sum_j q_j$ converges, then $j\, q_j \to 0$.*

Proof. (i) Convergence forces $q_j \to 0$. If $\rho_j \le M < 1$ for large j then $(1-M)^2 \le (1-\rho_j)^2 \le D_j \le (1+\rho_j)^2 \le (1+M)^2$, so D_j is bounded above and below by positive constants and $q_j \to 0$ is equivalent to its numerator $\rho_j(\rho_j + c_j) \to 0$. If $\rho_j \not\to 0$, pick a subsequence with $\rho_j \ge \delta > 0$; along it $\rho_j + c_j \to 0$. (ii) is Olivier's theorem applied to q_j, whose two hypotheses — eventual non-negativity, which by Corollary 13 is exactly $c_j \ge -\rho_j$, and eventual monotonicity — are both properties of the measured sequence itself. □

Proposition 18 recovers the initialization mechanism of Section 8 as a corollary with no computation: the pre-norm scaling $\rho_j = \kappa j^{-1/2}$ gives $\rho_j^2 = \kappa^2/j$, whose sum is harmonic and divergent, so $\sum_j q_j$ diverges. The closed form of Proposition 29 then supplies the *rate* of that divergence; the divergence itself needs neither the digamma function nor a fitted constant.

Remark 22 (What these hypotheses do on the measured sequences). Each proposition above is stated so that its hypotheses are properties of the measured $\{\rho_j\}, \{c_j\}$, checkable without a model. Checked on the pre-norm stack at initialization ($L = 32$, six seeds, the 63 bypassable branches of each), most of them fail. The alignment is not zero, so Proposition 18 does not apply. ρ_j is not monotone: it decreases at only 50–56% of consecutive steps, the $j^{-1/2}$ law of Section 8 being a trend through branch-to-branch scatter rather than a pointwise ordering, so Dirichlet's test and with it Proposition 20 do not apply. The partial sums C_N reach 1.8 to 11.7 over 63 branches, growing rather than visibly bounded. And 11 to 22 of the 63 coefficients are negative, so q_j is not eventually non-negative and Proposition 21(ii) does not apply either. What does survive is the comparison Proposition 18 would predict if it applied: measured at $L = 8, 16, 32, 64$, $\sum_j q_j$ and $\sum_j \rho_j^2$ both grow, with ratio $0.79, 0.66, 0.92, 0.56$ — persistently below the value 1 that zero alignment would force, which is a direct, unfitted measurement of what nonzero alignment removes from the sum. We report this as a negative result about the reach of the propositions, not as evidence for either answer.

The other quantity worth an exact statement is the deep-half average, because it is the one that looks empirically cleanest and is the most likely to be over-read.

Proposition 23 (Exact decomposition of the deep-half average)**.** *For attention sites $B(i) = 2(L-i+1)$ by (6). Write $e(i) := \bar{n}(i) - 1$ for the excess over the mandatory branch and $p_{\text{deep}} := \frac{2}{L}\sum_{i=1}^{\lfloor L/2 \rfloor} \bar{n}(i)/B(i)$. Then, exactly,*

$$L\, p_{\text{deep}} \;=\; \underbrace{\left(H_L - H_{\lceil L/2 \rceil}\right)}_{\to \log 2} \;+\; \sum_{i=1}^{\lfloor L/2 \rfloor} \frac{e(i)}{L-i+1}, \tag{11}$$

with H_n the n-th harmonic number ($H_0 := 0$). At the single deepest site the corresponding statement carries no error term at all: $L\, p_{L1} = \bar{n}(1)/2$ identically.

Proof. Substitute $B(i) = 2(L-i+1)$ and $\bar{n}(i) = 1 + e(i)$ into the definition and multiply by L; writing $m := \lfloor L/2 \rfloor$, the constant part is $\sum_{i=1}^{m} 1/(L-i+1)$, and the substitution $k = L-i+1$ sends $i = 1, \ldots, m$ to $k = L, \ldots, L-m+1$, so the sum equals $\sum_{k=L-m+1}^{L} 1/k = H_L - H_{L-m}$. Since $L - \lfloor L/2 \rfloor = \lceil L/2 \rceil$ for every integer L, this is $H_L - H_{\lceil L/2 \rceil} \to \log 2$; the two coincide when L is even and differ by exactly $1/\lceil L/2 \rceil$ when L is odd, which is why every depth in Table 4 being even leaves the numerical illustration below unaffected by which of the two is used. For the last claim, $B(1) = 2L$, so $L\, p_{L1} = L\, \bar{n}(1)/(2L)$. □

The first term of (11) is exactly the contribution of the mandatory branches, one per site, and it involves no property of the network whatever — it is $\log 2$ and arithmetic. Any empirical finding that $L\, p_{\text{deep}}$ is nearly constant is therefore, to leading order, a restatement of Corollary 9 and of the definition of the deep half. Section 8 quantifies how much: at the deepest ladder measured, the harmonic term supplies 0.685 of a measured 0.794, so 86% of the apparent law is the mandatory branch. Note also that $L\, p_{L1} = \bar{n}(1)/2$ and $L\, p_{\text{deep}} \to \log 2 + \cdots$ are different quantities and need not agree: the first reads one site whose cone is the whole stack, the second averages ratios over sites whose excesses $e(i)$ are small, and the measured values differ by about a factor of two for exactly that reason.

Remark 24 (What is and is not reduced). Theorem 12 proves something strong: the rest of the network enters q_j through exactly *two* scalars, their functional form is fixed, and both are observable at no cost beyond the two passes that give q_j. It does not eliminate the ordered product. In the sequential case ρ_j and c_j are defined through $U_{<j}$ and $U_{>j}d$, each a product of non-commuting factors, so while the two scalars are *measurable* they are not *predictable* from intrinsic quantities such as weight norms, layer index or architectural constants, and neither therefore is p. That is the obstruction, and it is an obstruction to prediction, not to computation. Two assumptions would lift it, each with a stated cost. *Transport neutrality:* if $U_{<j}$ acted conformally on $\operatorname{span}(K_j v_j, v_j)$ then ρ_j would reduce to the purely local $\|K_j v_j\|/\|v_j\|$, one vector–Jacobian product with no transport. But $U_{<j}$ is a product of the very operators whose non-conformality is the phenomenon under study — Proposition 27 makes that equivalence exact — and the distortion is largest at the deep sites where p is measured. *Alignment neglect:* setting $c_j = 0$ gives $q_j = \rho_j^2/(1+\rho_j^2)$, invertible as $\rho_j = \sqrt{q_j/(1-q_j)}$, which lies inside the bracket of Corollary 15 and may be substituted for the site-level $\sqrt{p/(1-p)}$ only under the further assumption that q_j is roughly uniform across the cone. But β_j is a function of σ_j, so c_j is not a generic cosine, and by Corollary 13 neglecting a systematically positive alignment *overestimates*

ρ_j when $\rho_j < 1$. The bias has a known sign, which makes the approximation usable as a bridge and unusable as an instrument. Corollaries 14 and 15 are what remain when neither assumption is granted, and they are exact.

5 The signs are not constrained

Write $w_k := \langle a_k, g\rangle/\|g\|^2$, so that $\sum_k w_k = \langle g, g\rangle/\|g\|^2 = 1$ and, by Theorem 4, $\bar{n} = \sum_k k\, w_k$. It is tempting to read $\bar{n}$ as a mean of the order k against a probability distribution $\{w_k\}$, and hence to conclude $0 \le \bar{n} \le B$ and $p \in [0, 1]$.

Proposition 25 (No sign constraint)**.** *The weights w_k and the coefficients q_j are not sign-constrained, and $\bar{n}$ is not bounded below. They are non-negative if the $\{a_k\}$ are mutually orthogonal, in which case $\{w_k\}$ is a probability distribution and $0 \le \bar{n} \le B$ follows; nothing in Lemma 2 supplies that orthogonality. Explicitly, for the one-branch site $g(\varepsilon) = \varepsilon(I + \varepsilon K)v$ with $Kv = -\lambda v$ and $\lambda \in (0, 1)$,*

$$q_{j_0} = 1, \qquad q_1 = w_1 = -\frac{\lambda}{1-\lambda}, \qquad \bar{n} = 1 - \frac{\lambda}{1-\lambda} \xrightarrow[\lambda \to 1^-]{} -\infty. \tag{12}$$

Proof. For the stated family, $a_1 = Kv = -\lambda v$ and a_0 vanishes at the site by Corollary 9, while $g = (1-\lambda)v$; hence $w_1 = \langle -\lambda v, (1-\lambda)v\rangle/\|(1-\lambda)v\|^2 = -\lambda/(1-\lambda)$, which is negative and diverges as $\lambda \to 1^-$, and $\bar{n} = q_{j_0} + q_1$ follows from (3). Equivalently this is Theorem 12 at $\rho_1 = \lambda$, $c_1 = -1$, the extreme point of Corollary 14. If instead $\langle a_k, a_l\rangle = 0$ for $k \ne l$ then $w_k = \|a_k\|^2/\|g\|^2 \ge 0$, which with $\sum_k w_k = 1$ gives the bounds. $\square$

Remark 26. Consequently $\bar{n} \ge 1$ is *false* in general, and does not follow from Corollary 9: that corollary gives $q_{j_0} = 1$, but $\bar{n} = 1 + \sum_{j \ne j_0} q_j$ and the remaining terms are not sign-constrained. Where $\bar{n} \ge 1$ and $p \in [0, 1]$ are observed they are measurements, and Section 7 reports that the negative part is not negligible. The mechanism is exactly the one (12) isolates: a branch whose contribution is anti-aligned with, and weaker than, the contribution routed around it. Any statement phrased as a "fraction of mass" must therefore carry the word *signed*.

6 Site dependence is exactly a change of metric

The coefficient q_j is indexed by a pair: the branch and the read site. The dependence is not arbitrary.

Proposition 27 (Transport identity)**.** *Let s and t be two read sites on one stream with t nearer the objective, so that $\mathrm{cone}(t) \subseteq \mathrm{cone}(s)$, and let T be the ordered product of the factors of (1) strictly between them, so that $g^{(s)} = Tg^{(t)}$ with all gates open; assume $g^{(t)} \ne 0$ and $g^{(s)} \ne 0$, as in Definition 3 at each site. Then for every $j \in \mathrm{cone}(t)$, with $M := T^{\mathsf{T}}T \succeq 0$ depending on the pair of sites but not on j,*

$$\beta_j^{(s)} = T\,\beta_j^{(t)}, \qquad q_j^{(s)} \;=\; \frac{\langle \beta_j^{(t)}, M\,g^{(t)}\rangle}{\langle g^{(t)}, M\,g^{(t)}\rangle}, \qquad \textit{versus} \qquad q_j^{(t)} \;=\; \frac{\langle \beta_j^{(t)}, g^{(t)}\rangle}{\langle g^{(t)}, g^{(t)}\rangle}, \tag{13}$$

so the whole site dependence is a change of inner product. Consequently the site-to-site difference is one linear functional evaluated at the branch parts,

$$q_j^{(s)} - q_j^{(t)} \;=\; \langle \beta_j^{(t)}, u_{s,t}\rangle, \qquad u_{s,t} \;:=\; \frac{M\,g^{(t)}}{\langle g^{(t)}, Mg^{(t)}\rangle} - \frac{g^{(t)}}{\|g^{(t)}\|^2}, \tag{14}$$

with $u_{s,t}$ depending on the pair of sites and not on j. Hence $q_j^{(s)} = q_j^{(t)}$ for every shared j if and only if $u_{s,t}$ is orthogonal to $\operatorname{span}\{\beta_j^{(t)} : j \in \operatorname{cone}(t)\}$; $u_{s,t} = 0$ if and only if $Mg^{(t)}$ is parallel to $g^{(t)}$, which holds in particular when T is conformal on that span; and the entire row $\{q_j^{(s)}\}_{j\in\operatorname{cone}(t)}$ is determined by the single vector $h_s := T^{\mathsf{T}} g^{(s)}$ through $q_j^{(s)} = \langle \beta_j^{(t)}, h_s\rangle / \langle g^{(t)}, h_s\rangle$.

Proof. For $j \in \operatorname{cone}(t)$ the factor of branch j lies to the right of T in (1), so T does not depend on ε_j and $g^{(s),(j\to 0)} = Tg^{(t),(j\to 0)}$; subtracting from $g^{(s)} = Tg^{(t)}$ gives $\beta_j^{(s)} = T\beta_j^{(t)}$. Then $q_j^{(s)} = \langle T\beta_j^{(t)}, Tg^{(t)}\rangle / \|Tg^{(t)}\|^2$, and moving T across the inner product gives (13). Subtracting the second display of (13) from the first and using bilinearity gives (14), from which the orthogonality criterion is immediate. The form (13) depends on M only through $Mg^{(t)} = T^{\mathsf{T}} g^{(s)} = h_s$, whence the last claim. □

Corollary 28 (The defect vector as a directional derivative)**.** *Let s, t be as in Proposition 27, and let y_s denote the residual-stream activation at the join immediately downstream of the branch defining site s (so $y_s = r_i$ or $y_s = x_i$ in the notation of Section 1, according to which of a layer's two joins s is), with y_t defined the same way for t. Let $\Phi_{s\to t}$ denote the network's own forward map from y_s to y_t — the ordinary composition of the joins strictly between them, at the fixed input producing $g^{(s)}$ and $g^{(t)}$, and independent of every ε_j by Definition 1. Then*

$$T \;=\; J_{\Phi_{s\to t}}(y_s)^{\mathsf{T}}, \qquad \text{so} \qquad h_s \;=\; T^{\mathsf{T}} g^{(s)} \;=\; J_{\Phi_{s\to t}}(y_s)\, g^{(s)}, \tag{15}$$

the directional derivative of $\Phi_{s\to t}$ at y_s in the direction $g^{(s)}$. Consequently, for $u := g^{(s)}/\|g^{(s)}\|$ and $\varepsilon > 0$ small enough that $\Phi_{s\to t}$ is three-times continuously differentiable on the segment $[y_s - \varepsilon u,\, y_s + \varepsilon u]$,

$$\left\| \|g^{(s)}\| \cdot \frac{\Phi_{s\to t}(y_s + \varepsilon u) - \Phi_{s\to t}(y_s - \varepsilon u)}{2\varepsilon} \;-\; h_s \right\| \;\le\; \frac{\|g^{(s)}\|\,\varepsilon^2}{6} \sup_{\xi\in[y_s-\varepsilon u,\, y_s+\varepsilon u]} \left\| D^3\Phi_{s\to t}(\xi)\right\|. \tag{16}$$

Proof. By Definition 1, a bypassable join contributes the factor $I + \varepsilon_j K_j$ to (1) with $K_j = J_{f_j}^{\mathsf{T}}$, so at $\varepsilon_j = 1$ that factor is $(I + J_{f_j})^{\mathsf{T}}$, the transpose of the local join's own forward Jacobian $I + J_{f_j}$; a join every path traverses contributes $C_j = J_{f_j}^{\mathsf{T}}$ directly, likewise a transposed local forward Jacobian. T is the ordered product of exactly these factors for the joins strictly between s and t, by Proposition 27's own proof, hence the transpose of the ordered product, in reverse order, of the corresponding untransposed local Jacobians — which is $J_{\Phi_{s\to t}}(y_s)$ by the chain rule, since $\Phi_{s\to t}$ is exactly the composition of those same joins as ordinary forward maps and $y = x + f(x)$ never depends on ε (Definition 1). This gives (15). Equation (16) is the standard remainder bound for a centered difference: expanding $\Phi_{s\to t}(y_s \pm \varepsilon u)$ to third order about y_s, the zeroth- and second-order terms cancel in the difference, the first-order term is exactly $2\varepsilon\, J_{\Phi_{s\to t}}(y_s)\, u$, and the third-order remainder is bounded termwise by the stated supremum; dividing by 2ε and scaling by $\|g^{(s)}\|$ gives the claim. □

Proposition 27 says what may and may not be done with a per-site profile. It may not be differenced to recover per-branch coefficients, since the branches shared by two sites carry different coefficients there. But the deviation is not free structure: it measures the non-conformality of the transport, and it is constrained, since one vector per site generates a whole row.

Statement	Identity checked	N	Deviation
Cor. 8	$q_j = 0$ upstream of the site	1540	1.06×10^{-7}
Cor. 9	$q_{j_0} = 1$ at the site	56	0
Thm. 12	$q_j = q(\rho_j, c_j)$	1596	2.50×10^{-16}
Cor. 13	$\operatorname{sign} q_j = \operatorname{sign}(\rho_j + c_j)$	1596	0 violations
Cor. 13	$q_j < 0 \Rightarrow c_j < -\rho_j,\ \rho_j < 1$	258	0 violations
Thm. 6	$\bar{n}$ per layer vs. central difference	7	≤ 0.0098

Table 1: Verification on the trained model. The upstream residual sits at the single-precision floor and is the substantive gate, per Remark 11. The last row compares the exact identity against the finite-difference estimator of Definition 3 at $h = 0.05$: p over the deep half is 0.16584 exactly against 0.16590 by difference, with a maximum per-layer deviation of 0.0098 in $\bar{n}$ and a median of 0.0029. The estimator is therefore accurate to about one part in a thousand on p, and Theorem 6 confirms rather than corrects it.

One caution about its empirical status, since the two gates of Section 3 invite the comparison. Corollary 8 is over-determined — it predicts a specific numerical value, zero, that a wrong implementation would miss — whereas (14) at a single pair of sites is not: the unknown $u_{s,t}$ lives in the readout's ambient dimension, which exceeds the cone size by orders of magnitude, so a least-squares u fits any observed deviations exactly. Equation (14) is therefore offered as structure and as a constraint on how a per-site profile may be interpreted, and it is verified as algebra in Section 7; it is not a test that a network can fail, and we do not present it as one. Corollary 28 is a different kind of check on the same equation: it does not fit $u_{s,t}$ to the observed deviations at all, but predicts them from an independent forward computation that never sees $q_j^{(s)}$ or $q_j^{(t)}$, and this one a network can fail.

7 Numerical verification

The identities were verified on a trained 1.5B-parameter pre-norm Transformer ($L = 28$, 12 heads, grouped-query attention, $d_{\text{model}} = 1536$), seeded from a next-token cross-entropy on one tokenized prompt of 16 tokens, with the gate of Definition 1 inserted on all $2L = 56$ branches. That the forward pass is untouched — the hypothesis of Lemma 2 — was confirmed bit-exactly: the logits and all retained parameter gradients at $\varepsilon \equiv 1$ are identical to those of the ungated graph in the last bit. A two-sided control confirms the skip path survives: closing all gates zeroes every in-block normalization gradient exactly, while the final normalization gradient, which lies outside every branch, remains non-zero.

Table 1 collects the checks on the input first measured; the finite-difference comparison in its last row exists only for that one input, since it is a check on the estimator of Definition 3 rather than on the network, but every other row was re-checked on the full 22-prompt sweep used below. The worst case across prompts is 4.07×10^{-7} for Cor. 8 (row 1, up from 1.06×10^{-7} on the single input) and 1.64×10^{-15} for Thm. 12 (row 3, up from 2.50×10^{-16}) — both grow with the sweep but stay at the double-precision noise floor for a computation of this depth, and the sign law (rows 4-5) records zero violations on every one of the 35,112 coefficients in the sweep. The deviations in this table are therefore representative rather than a favourable single case: the values shift with the input but the floor they sit on does not. The value of Theorem 6 is accordingly not a numerical correction to $\bar{n}$: it is that the per-branch coefficients become

layer i	$L-i$	B	$\bar{n}$ median	$\bar{n}$ range	p median (range)
1	27	56	6.322	[3.368, 9.458]	0.113 [0.060, 0.169]
2	26	54	5.700	[2.671, 8.994]	0.106 [0.049, 0.167]
5	23	48	5.370	[3.987, 7.597]	0.112 [0.083, 0.158]
12	16	34	4.299	[2.769, 5.978]	0.126 [0.081, 0.176]
16	12	26	3.718	[2.372, 5.089]	0.143 [0.091, 0.196]
22	6	14	2.326	[1.432, 3.079]	0.166 [0.102, 0.220]
27	1	4	1.313	[1.113, 1.892]	0.328 [0.278, 0.473]
28	0	2	1.123	[0.944, 1.517]	0.562 [0.472, 0.758]

Table 2: Exact $\bar{n}$ and p at attention sites of the trained model, over 22 *input prompts*, abridged from the 28-row profile. Median and full range across prompts; the spread is the point, and reporting a single input's column would misrepresent it. $\bar{n}$ falls monotonically towards the objective at the median while p rises, because B falls faster. Averaged over layers 1–14, p has median 0.1161 and range [0.0778, 0.1658], a coefficient of variation of 19%. Every entry is the exact identity of Theorem 6, not an estimate.

observable at all. No finite-difference scheme at any h yields q_j, and the sign law, the envelope, the bracket and the site-dependence measurement below all descend from the ablation identity rather than from Definition 3. That two independent estimators of $\bar{n}$ agree to 3×10^{-4} is mutual corroboration of two implementations.

Table 2 reports the profile across 22 prompts. At the layer-1 site the attainable order is $B = 56$ and the median measured order is $\bar{n} = 6.32$, ranging from 3.37 to 9.46 by input, so the readout's mass sits at a small fraction of the available orders, yet far above the shortest paths: it is not carried by order-one paths alone. The contrast with the raw path count is stark. In a configuration of the same architecture family with $L = 24$ layers and $H = 16$ attention heads (unrelated to the harmonic number H_n of Proposition 23), the exact number of directed graph paths from a layer-1 attention site to the output is $3r^{L-1}$ with $r(H) = 6H^3 + 3H^2 + 3$, about 10^{101}, verified against the graph with zero residual by the enumeration of [2], while the branch order at that site cannot exceed $2L = 48$. Grouping paths by branch order is what makes the question tractable at all.

Proposition 25 is not vacuous at this scale. Across the 22 prompts, each contributing 1596 in-cone coefficients for 35,112 in all, the negative fraction has median 12.9% and ranges from 4.2% to 24.4% by input. Assuming the orthogonality of Proposition 25 is therefore not a harmless simplification on trained weights: it is false on every input tested. Proposition 25 also predicts that $\bar{n} \geq 1$ can fail, and on this model it does: at the layer-28 site, whose cone is its own gate and the layer-28 MLP gate, the minimum over prompts is $\bar{n} = 0.944$. That is a measured readout whose mean branch order is below the mandatory branch's own contribution of 1, which the constructed family (12) anticipated and which no positivity-based reading of $\bar{n}$ can accommodate.

The sign law is confirmed at this scale as well, and sharply: on every one of the 22 prompts the fraction of branches with $c_j < -\rho_j$ equals the fraction with $q_j < 0$ to within 0.000 percentage points, and Corollary 13 records *zero* violations across all 35,112 pairs.

That equality is not a coincidence but Corollary 13 doing work, and it locates where the negative coefficients come from. Since $q_j < 0$ if and only if $c_j < -\rho_j$, negative coefficients require negative alignment, and on this trained model the alignment is negatively biased: at the layer-1 site the mean of c_j over the 55 bypassable branches is negative on 21 of the 22 prompts, with mean -0.100 and median -0.090 across prompts, against $+0.037$ with 56% of values positive on the synthetic pre-norm stack at initialization. The one exception, at $+0.029$, matters: the bias is very much the majority behaviour but it is not universal, and we report it as the former. The amplitude is strongly input-dependent, with a coefficient of variation of 62%, so what generalizes here is the sign and not the size. A negative correlation between prompt length and mean alignment that appeared on a ten-prompt subset ($r = -0.55$) fell to $r = -0.217$ on all 22 and should be read as a small-sample artifact; the length correlation that survives is with p_{L1} itself, at $r = +0.556$.

Site dependence, per Proposition 27, is modest in the middle of the distribution and severe in the tail, and this is now measured across all 22 prompts rather than one: 1540 pairs of adjacent sites sharing a branch, times 22 inputs, 33,880 pairs in all. The symmetric deviation $|q^{(i)} - q^{(i+1)}|/(|q^{(i)}| + |q^{(i+1)}|)$ has median 0.027 and ninetieth percentile 0.262, with 6.4% of pairs of opposite sign; in absolute terms the median deviation is 0.0033 against a median $|q_j|$ of 0.078, or 4.2% of that scale at the median and 30% at the ninetieth percentile. Every one of these is at or above its single-prompt counterpart (0.019 median, 0.203 ninetieth percentile, 4.9% opposite sign): the earlier measurement was not merely small in sample, it sat on the mild side of what a single input happened to show. This shape is what (14) leads one to expect: the deviations across branches are not independent quantities but the values of a single linear functional $\langle \cdot, u_{s,t} \rangle$ on the branch contributions, so in high dimension most β_j are nearly orthogonal to $u_{s,t}$ and give a small deviation while the few aligned with it give a large one. The transport is close to conformal on most of the cone and badly non-conformal on a little of it, and that tail is heavier than the single-input measurement showed.

Corollary 28 was checked on the production model, not only as algebra. Two implementation details mattered and are reported for reproducibility: an ordinary graph re-demand of a downstream node after overwriting an upstream one is not enough on its own to force recomputation in this framework, since the perturbed node keeps its original computation rule and can be silently recomputed from its true ancestors rather than from the overwritten value, so that rule must be suspended for the perturbed node during the measurement and restored after; and y_s must be the residual-stream join output, not the raw branch output, since the latter omits exactly the skip term that carries the site's own identity contribution downstream. With both corrected, on the production checkpoint, s at the layer-1 attention site and t at the layer-13 attention site ($|\mathrm{cone}(t)| = 32$), the centered difference of (16) at $\varepsilon = 10^{-2}$ predicts the 32 directly and independently measured values $q_j^{(s)}$ — via $\hat{q}_j^{(s)} := \langle \beta_j^{(t)}, \hat{h}_s \rangle / \langle g^{(t)}, \hat{h}_s \rangle$ — to a median absolute error of 1.10×10^{-5}, against a median $|q_j^{(s)}|$ of 8.2×10^{-2}: five significant figures of agreement between a quantity built entirely from an independent forward computation and one measured entirely by backward ablation. The error grows rather than shrinks at smaller ε (1.28×10^{-4} at $\varepsilon = 10^{-3}$, 1.07×10^{-3} at $\varepsilon = 10^{-4}$), an empirical slope near -1 rather than the $+2$ of (16): this is the round-off half of the truncation/round-off trade-off standard to any centered difference, not a failure of the bound, and it is checkable quantitatively rather than merely plausible. Modelling the round-off term alone as C_2/ε, the three measurements give $C_2 = \varepsilon \times \text{error} \in \{1.10, 1.28, 1.07\} \times 10^{-7}$, consistent to within 15% across these three step sizes and centred on 1.15×10^{-7} — essentially the single-precision unit round-off, 1.19×10^{-7}, which is exactly what cancellation in $\Phi_{s\to t}(y_s + \varepsilon u) - \Phi_{s\to t}(y_s - \varepsilon u)$ predicts. This tightness is a property of one instance probed at three step sizes, not of C_2 across instances: holding $\varepsilon = 10^{-2}$ fixed and repeating the measurement on six further, independent prompts at the same site pair gives C_2 ranging over a factor of 4.25 (from 5.28×10^{-8} to 2.25×10^{-7}), with the seven-prompt median, 1.10×10^{-7}, still within 8% of the single-precision floor. Since the error decreases monotonically as ε grows across the whole tested range, the optimum of (16)'s trade-off lies at or above $\varepsilon = 10^{-2}$; seeing the bound's own $+2$ slope would need an order of magnitude or two above that, which was not tested. One coefficient out of the 32 keeps a fixed error of exactly 1.0020 at every ε tested, and this is not a numerical artifact but an exact algebraic consequence, independent of ε and of $\hat{h}_s$'s accuracy: a site captured after its own gate satisfies Corollary 9, so at $j = t$, $\sigma_t^{(t)} = 0$ and $\beta_t^{(t)} = g^{(t)}$ exactly, and substituting into the prediction formula gives $\hat{q}_t^{(s)} = \langle g^{(t)}, \hat{h}_s\rangle / \langle g^{(t)}, \hat{h}_s\rangle = 1$ for every $\hat{h}_s$ with $\langle g^{(t)}, \hat{h}_s\rangle \neq 0$, at any ε. The fixed error is therefore exactly $|1 - q_t^{(s)}|$, a property of the independently measured $q_t^{(s)} \approx -0.002$ and not of the finite-difference estimator at all.

Independently, and in double precision, the decomposition was checked against ground truth obtained without any identity: on a synthetic instance with deliberately non-normal, non-commuting factors and $B = 12$, the coefficients a_k were computed by exhaustive expansion of (1) over all 2^{12} subsets. Against that reference the ablation identity (5) reproduces $\bar{n}$ to 4×10^{-16} and the two-scalar form (7) to 1×10^{-16} per coefficient, while the central difference of Remark 7 at $h = 0.05$ deviates by 1.3×10^{-3}. In the same instance $\sum_k w_k = 1$ to twelve digits while $w_6 = -1.0 \times 10^{-2}$: the order weights of Section 5 are negative by a percent of the total, not by a rounding error. On a synthetic pre-norm stack of the form of Section 1, Corollaries 13, 14 and 15 record zero violations over 186 coefficients, the transport identity (13) and the defect identity (14) each hold to 2×10^{-16}, a conformal transport gives site invariance to 3×10^{-16}, and the family (12) reproduces $\bar{n} = -8$ at $\lambda = 0.9$ and $\bar{n} = -98$ at $\lambda = 0.99$.

A second, independently seeded exhaustive check, with $B = 16$ and per-branch rather than only aggregate ground truth, was built to exercise two regimes the $B = 12$ instance did not report: branches with $\rho_j > 1$ and branches at the envelope's extreme $c_j = -1$. Two branches were constructed with $\rho_j > 1$ ($\rho_j = 1.0507$ and $\rho_j = 1.2053$), both giving $q_j > \frac{1}{2}$ as the threshold of Corollary 14 requires; two reflection branches $-\lambda\, vv^\top/(v^\top v)$ were constructed with $\rho_j = \lambda$ exactly and $c_j = -1$ exactly, where the corollary's lower bound is attained. At $\lambda = 0.85$ the measured $q_j = -5.66666667$ against the closed form $-\lambda/(1-\lambda) = -5.66666667$, and at $\lambda = 0.40$, $q_j = -0.66666667$ against -0.66666667, both to machine precision: the envelope is not merely an inequality that happens to hold but is attained exactly where the construction places it. Over the twelve bypassable branches of the cone in this instance, 2 have $\rho_j > 1$ and 5 have $q_j < 0$; the sign law and the envelope of Corollary 14 hold on all twelve with zero violations, so the check exercises both sides of the threshold rather than only the trivial one.

8 Depth: what a trained ladder settles, and what it does not

The decomposition invites a question it does not answer: how does $\bar{n} = \sum_j q_j$ grow with the number of branches in the cone? A scaling argument suggests strongly sub-linear growth in a pre-norm stack, and it is not a theorem. Pre-normalization fixes the scale of each branch's *input*, so branch outputs have depth-independent scale, while the residual stream accumulates them incoherently and $\|x_j\|$ grows like $\sqrt{j}$. The branch Jacobian carries the normalization factor $1/\mathrm{RMS}(x_{j-1})$, so ρ_j should decay like $j^{-1/2}$; with c_j small this gives $q_j \approx \rho_j^2/(1 + \rho_j^2) \sim 1/j$ by Corollary 14, hence $\bar{n} \sim \log L$ and $p \sim \log L/L$.

Table 3 tests the argument on a synthetic pre-norm stack with random weights, where the predicted decay is visible directly: the product $\rho_j\sqrt{j}$ stays within 0.79 and 1.15 across $j = 2, \ldots, 64$ at $L = 32$. Logarithmic growth fits better than the best power law on the same target and criterion, and linear growth is excluded by a factor of twelve. The comparison to make is

L	B	$\bar{n}$ (median)	p (median)	$\bar{n}/\log L$
4	8	2.1434	0.26793	1.546
8	16	2.8359	0.17724	1.364
16	32	3.3300	0.10406	1.201
32	64	4.6180	0.07216	1.333
64	128	5.2767	0.04122	1.269
128	256	5.4320	0.02122	1.120

Table 3: Synthetic pre-norm stack at *random initialization*, 8 seeds per depth, exact coefficients by Theorem 6 in double precision. Fitting $\bar{n}$ against the same target by least squares, $\bar{n} = 0.718 + 1.033 \log L$ attains $R^2 = 0.965$ while the best power law $1.722\, L^{0.252}$ attains $R^2 = 0.929$. Across this 32-fold range of depth, linear growth would require $\bar{n}$ to rise by a factor 32; the measured factor is 2.53 and the fitted logarithmic form predicts 2.66.

against the *fitted* form, which carries an intercept: $0.718 + 1.033 \log L$ predicts a rise of 2.66 against the measured 2.53, an agreement of five percent. Comparing instead against a pure $\log L$ with no intercept would predict 3.50 and manufacture a shortfall that the data do not contain. What this range cannot do is separate $\log L$ from a slowly growing power such as $L^{0.25}$; it separates both from linear, which is the question at issue.

The fitted intercept 0.718 has, so far, no content beyond being the value that makes the line pass through the points. The idealized model of the opening paragraph is precise enough to do better: taken literally, with no further assumption, it determines the whole excess sum in closed form.

Proposition 29 (Exact excess sum under the idealized model)**.** *Assume the model of the opening paragraph exactly: $c_j \equiv 0$ (alignment neglect) and $\rho_j = \kappa/\sqrt{j}$ for a constant $\kappa > 0$, for $j = 1, \dots, N$ bypassable branches. Then $q_j = \kappa^2/(j + \kappa^2)$ exactly by Corollary 14 at $c = 0$, and*

$$e(N) \;:=\; \sum_{j=1}^{N} q_j \;=\; \kappa^2 \big[\psi(N + 1 + \kappa^2) - \psi(1 + \kappa^2)\big], \tag{17}$$

with ψ the digamma function. As depth $L \to \infty$, with $N = 2L - 1$ bypassable branches,

$$e(L) \;=\; \kappa^2 \log L \;+\; \Big[\kappa^2 \log 2 - \kappa^2 \psi(1 + \kappa^2)\Big] \;+\; \frac{\kappa^2(2\kappa^2 - 1)}{4L} \;+\; O(L^{-2}). \tag{18}$$

Proof. The recurrence $\psi(x+1) = \psi(x) + 1/x$ gives $\sum_{j=1}^{N} \frac{a}{j+a} = a[\psi(N+1+a) - \psi(1+a)]$ for any a; setting $a = \kappa^2$ gives (17). Substituting $N + 1 + \kappa^2 = 2L + \kappa^2$ and the standard large-argument asymptotic expansion $\psi(z) = \log z - \frac{1}{2z} - \frac{1}{12z^2} + O(z^{-4})$ [5] at $z = 2L + \kappa^2$, then expanding $\log(2L + \kappa^2) = \log 2 + \log L + \kappa^2/(2L) + O(L^{-2})$ and collecting powers of $1/L$, gives (18); the term $\psi(1 + \kappa^2)$ is left unexpanded, since $1 + \kappa^2$ is a fixed constant, not a large argument. □

Equation (17) is exact given the model, not a fit: it has one free parameter, the same κ the model already has, and no others. Fitting that one parameter to the six medians of Table 3 by least squares gives $\kappa^2 = 0.864$ ($R^2 = 0.93$), reproducing the empirical log-slope 1.033 to within 16% but falling short of the two-parameter fit's $R^2 = 0.965$; its closed-form intercept, $\kappa^2 \log 2 - \kappa^2 \psi(1 + \kappa^2) = 0.31$, is well below the fitted 0.718. The idealized model does not close the gap, and we report that rather than the flattering half of it.

Remark 30 (Why the gap is not a surprise, and what it bounds). Expanding the two-scalar form (7) for small ρ at general (not necessarily zero) alignment c gives $q(\rho, c) = \rho c + \rho^2(1-2c^2) + O(\rho^3)$: the alignment term that Proposition 29 sets to zero enters *linearly* in ρ, one order below the quadratic term the proposition keeps. With $\rho_j = \kappa/\sqrt{j}$, a persistent nonzero mean alignment would therefore contribute $\Theta(\kappa\bar{c}\sqrt{L})$ to the excess — asymptotically faster than the $\kappa^2 \log L$ of (18) — so whether the excess stays logarithmic or crosses over to $\sqrt{L}$ growth turns on whether c_j carries a persistent sign as j grows, not on whether any individual c_j vanishes: the condition the logarithmic account actually needs is that $\sum_{j\leq N} c_j$ not grow like $\sqrt{N}$, which is weaker than, and different from, "$c_j \approx 0$." Measured directly on the same synthetic model at $L = 32$ (six seeds, the 63 bypassable branches of each), c_j does not vanish per branch — it ranges from -0.672 to 0.784 — but its mean over all 378 measured values is 0.037, an order of magnitude smaller than its spread, with 56% of values positive: individually large, collectively close to unbiased. Consistently, a one-parameter fit of pure $\sqrt{L}$ growth to the six medians of Table 3 attains only $R^2 = 0.74$, markedly worse than Proposition 29's own 0.93. Over the depths tested, the near-zero mean is doing the work the pointwise assumption cannot: the individual c_j are not small, but their sum does not accumulate the way a persistent sign would require. We did not verify the sharper $\sqrt{N}$-growth condition directly — only its two available proxies, the mean and the $\sqrt{L}$ exclusion — and name the direct check as the natural next step rather than a settled matter.

Both the proposition and the remark are statements about the idealized model at initialization, verified against the same synthetic sweep the model was proposed for; nothing here is evaluated on, or claimed for, a trained network, and Table 4 below shows the mechanism itself does not survive training in any case.

A ladder trained at fixed width

To ask the same question of trained models one needs several depths at *matched* width and matched training. A family of production models cannot supply it. Depth in such a family is tied to memory, and the engine used here is Float32-only, so a model costs four bytes per parameter for weights alone before any activation: the 3 B member of the family at $L = 36$ needs some 12 GB before it begins to run, and the accelerator available here holds the 1.5 B model at $L = 28$ — on which every Qwen measurement above was taken — but not that one. The attainable trained depths are therefore $L = 24$ and $L = 28$, a range of 1.17, against the 32-fold range Table 3 needed to separate the logarithmic form ($R^2 = 0.965$) from a power law (0.929), where the margin was already modest. Two depths spanning 1.17 distinguish nothing, whatever the hardware. And in every production family width co-varies with depth, so even a successful sweep would confound the two.

We therefore trained the ladder: $L = 4, 8, 16, 32, 64$ at fixed $d_{\text{model}} = 384$ and hidden dimension 1024, three seeds each, character-level on a 1.1 MB corpus with a 90/10 split, sequence length 64 — inside the range where p is flat in n, so length cannot confound depth. Holding width fixed removes the width confound by construction, which no off-the-shelf family can do. What is traded away must be said plainly: a ladder one can afford to train is small and character-level, so the claim is "at fixed width, on models trained for this purpose" — weaker in realism than a production family, stronger in causal identification. We chose that trade deliberately. The protocol was fixed before the runs: the control for training amount is *matched validation loss* rather than matched steps or matched FLOPs, because matched steps under-trains the deeper models and so biases towards the very logarithmic form under test, and matched FLOPs biases the same way more strongly; each depth must reach a common loss $\tau = 2.60$ within the budget

	at initialization			trained, matched loss		
L	$\bar{n}$	$e/\log L$	% neg.	$\bar{n}$	$e/\log L$	% neg.
4	2.9788	1.427	0.0	2.2730	0.918	5.6
8	3.7212	1.309	0.0	2.5067	0.725	12.5
16	4.6149	1.304	0.0	2.6929	0.611	21.0
32	5.7382	1.367	0.8	2.6429	0.474	30.3
64	6.3120	1.277	2.9	3.0006	0.481	24.4

Table 4: The same ladder — same architecture, widths, seeds, read site and input — measured with zero training steps and after training to a common validation loss. Medians over three seeds; $e := \bar{n} - 1$ is the excess over the mandatory branch of Corollary 9, and $e/\log L$ is flat if and only if the per-branch coefficients decay like $1/j$. At initialization it is flat (1.28 to 1.43) and the excess grows by 2.68 against the 3.00 a logarithmic sum predicts. Trained, it falls by half and the excess grows by only 1.57. The negative fraction is measured over all in-cone pairs, 36 at $L = 4$ and 8256 at $L = 64$.

and beat a unigram baseline, with failures reported rather than dropped; the read site is the layer-one attention site, whose cone is the whole stack; and three seeds per depth are reported by their median. All fifteen models passed the gate.

Table 4 isolates training. Both regimes are measured on the same architecture, the same widths, the same seeds, the same read site and the same input tensor, with nothing differing but the optimizer steps — which matters, because the synthetic stack of Table 3 differs from a trained transformer in more than training, and attributing the contrast to training alone would otherwise confound training with architecture. Three things follow.

First, the mechanism of the opening paragraph is confirmed *at initialization on a real transformer*, not merely on a synthetic stack: $e/\log L$ is flat between 1.28 and 1.43 across a 16-fold range, which is what $q_j \sim 1/j$ predicts and what makes $\sum_j q_j$ diverge logarithmically.

Second, that mechanism does not survive training. Trained at matched loss, $e/\log L$ falls from 0.918 to 0.481, so the coefficients decay faster than $1/j$ and the growth of $\bar{n}$ is sub-logarithmic. Depth behaviour is therefore a property of *training*, not of the architecture — which is a stronger and more interesting statement than the one the initialization data alone supports.

Third, cancellation is created by training as well. At initialization the coefficients are essentially all positive: no negatives at all up to $L = 16$, 2.9% at $L = 64$, and a cancellation index $\sum_j |q_j|/|\sum_j q_j|$ of 1.000 to 1.026. Trained, the negative fraction rises with depth to between 24% and 30% at $L = 32$ and 64, against a median 12.9% over 22 inputs on the production model at $L = 28$. So $\bar{n}$ is a genuine positively weighted mean of orders at initialization and becomes a signed quasi-mean only once trained: Proposition 25 is not merely unfalsified, it describes the trained regime specifically.

What is not settled: convergence. We cannot say whether $\sum_j q_j$ converges. The excess still grows over the measured range — by 1.57 at the median, and by at least 1.25 taking the worst pairing of seed extremes — so convergence *within* $L \leq 64$ is excluded, and convergence beyond it is not. Fitting the excess as a power of the cone size gives exponent 0.143, i.e. $q_j \sim j^{-0.86}$, which diverges; a convergent alternative with $q_j \sim j^{-1.1}$ predicts a rise of 1.76 over the same range against the measured 1.57. Five depths and three seeds cannot separate a slowly divergent sum from a convergent one approached slowly, and we do not claim to. What Theorem 17 contributes is the criterion that would settle it, in a quantity already measured: boundedness of $\sum_j \rho_j/(1-\rho_j)$. Since the negative fraction grows with depth, a limit claim would additionally need the cancellation index bounded, not merely $\sum_j q_j$ convergent — Theorem 17 is a statement about absolute convergence for exactly that reason. That index is reported here at initialization only; measuring it across the trained ladder is the next step and costs nothing beyond the $B+1$ passes already required.

The confound that remains. Matching at a common loss does not match how converged each model is relative to its own potential. Median final loss after the full budget was 2.17 at $L = 4$ against 2.52 at $L = 64$: the deeper models train more slowly per step at fixed learning rate and batch size one, reach τ late, and improve little afterwards. Every trained-depth statement above carries that. Its direction is known and it works against us in one place and for us in another: under-training the deep models suppresses their $\bar{n}$ and so favours the sub-logarithmic reading, which is the reading we report; whereas the same effect cannot explain the initialization-versus-trained contrast, since the initialization arm involves no training at all. The sub-logarithmic finding should be read as provisional on that ground, and the initialization-to-trained contrast as robust to it. We looked for a magnitude, not only a direction, using the loss–$\bar{n}$ pairs already in hand at each depth (the matched-loss and fixed-step measurements of the same three seeds, two points per seed): across all 15 seed–depth combinations the within-depth slope of $\bar{n}$ against loss has the expected sign on 13, and the wrong one on 2 — both at the two deepest depths, $L = 32$ and $L = 64$, one seed each, where $\bar{n}$ rises as loss falls. Two points per seed cannot support a defensible extrapolation even where the sign is consistent, and the inconsistency is concentrated exactly where a magnitude estimate would matter most, so we report the direction and no magnitude; manufacturing one would be worse than leaving it disclosed.

Finally, Proposition 23 explains an apparent regularity that would otherwise be over-read. On the trained ladder, $L\,p_{\text{deep}}$ is nearly constant — $0.924, 0.874, 0.812, 0.778, 0.794$ across a 16-fold range of depth, a coefficient of variation of 6.5% — which invites the reading that $p \propto 1/L$ is a law. Most of it is not a law about the network. At $L = 64$ the harmonic term $H_{64} - H_{32} = 0.685$ accounts for 86% of the measured 0.794, and that term is exactly the mandatory branch of Corollary 9 summed over sites: it would take the same value in any network with this join structure. The network-dependent remainder is the second sum in (11), which measures $0.340, 0.240, 0.149, 0.100, 0.109$ — small, and falling with depth. The same quantity at initialization is not constant at all, $L\,p_{\text{deep}}$ rising from 1.411 to 1.804, which is what (11) predicts when the excesses are large. The near constancy is thus a trained-regime coincidence between a fixed arithmetic term and a shrinking one, and we report it as such rather than as a scaling law.

9 Scope

Lemma 2, Theorems 4, 6, 12 and 16, Corollaries 8–15, Proposition 25 and Proposition 27 are architecture-free: they hold for any multi-affine G of the form (1), hence for any additively joined branch structure, with (6) additionally requiring that the branch order be total. The measurements are not architecture-free, and five boundaries deserve emphasis.

First, the paper's primary limitation, stated once here in full and not softened. Two different things are true of the production-model numbers, and they should not be conflated; the depth ladder of Section 8, by contrast, is a separate experiment with three seeds per depth and is not affected by what follows. The exact identity has now been applied to all 22 prompts, so the per-layer profile of Table 2, the negative fraction, the alignment statistics and the sign-law verification rest on 35,112 coefficients across 22 inputs rather than on one. What that closed was not a question of accuracy but of representativeness, and the answer matters: the single input previously reported is prompt 6 of the 22, and it is not typical. Its $\bar{n} = 8.97$ at the layer-1 site sits near the top of a distribution whose median is 6.32, and its $p_{L1} = 0.160$ against a median of 0.113. The re-measurement reproduces that prompt's published values exactly, to the last printed digit, so nothing was wrong with the earlier number; it was simply one draw, and an atypical one.

The site-dependence quantiles are recomputed across the same 22 prompts below as well, from the same coefficient matrix, at no further measurement cost; every quantity in this section is now cross-input. The coefficients depend on the seed d by construction, so p is strictly a property of the quintuple (network, branch partition, site, seed, target); across the 22 prompts the branch fraction at the layer-1 site has a coefficient of variation of 25% and the mean alignment 62%, and nothing here explains *why* either varies.

Second, p is a property of the pair (network, branch partition), not of the network: refining the gates to per-head granularity changes both $\bar{n}$, since a path crossing three heads of one layer then counts three traversals, and B, and we do not claim to know the sign of the net effect. The partition must be fixed in the definition, and p is comparable only at fixed partition.

Third, block topology changes the object rather than its value. A parallel block yields two gates per layer that no single path can both traverse, so cone size and maximum order diverge as in Remark 10, and p is not comparable across topologies. The measurements assume two sequential joins per layer and no skip connection bypassing a whole layer.

Fourth, q_j depends on the read site, so p is a per-site quantity and a per-site profile may not be differenced to recover per-branch coefficients; Proposition 27 says precisely how much freedom the site dependence has, and the measurement above says it is used.

Fifth, $\bar{n} \geq 1$, $\bar{n} \leq B$ and $p \in [0, 1]$ are measurements, not consequences of the construction; Proposition 25 exhibits readouts violating the first two, a median 12.9% of the measured coefficients are negative on every one of 22 inputs, and $\bar{n} = 0.944 < 1$ is observed directly at the layer-28 site.

We add for completeness that p is close to flat in sequence length on the trained model, $p = 0.1243 \pm 0.0048$ for $n \geq 16$ over a 32-fold range, with the best of three fitted forms reaching only $R^2 = 0.58$; we report this as a measured flatness and offer no account of it here.

Remark 31 (What this note does not claim). A separate line of work bounds the loss of dependence on the seed d — target-independence — in *non-negative* modified-backpropagation schemes, where products of positive maps contract towards a rank-one attractor [4]. The factors here are signed, and $\bar{n}$ measures the order of gradient mass rather than the sensitivity of g to d; these are different functionals. Any bridge between them is a conjecture, and the experiment that would establish it is explicit: measure the cosine between readouts seeded from two distinct targets, across sites and depths, and test whether it tracks $\bar{n}$ rather than depth. Likewise, the observation that residual networks behave like ensembles of shallow paths [3] is consistent with $\bar{n} \ll B$ but not with $\bar{n}$ bounded: at $L = 28$ the measured order has median 6.32 and reaches 9.46 on some inputs, an order of magnitude above what a short-paths-only account would predict. On the abstract's estimand caveat, a companion paper documents a concrete instance of the same dissociation, by a different route: a grafted branch's gate and gradients there stay at *exact* zero for the entire 600-step run reported while the host network's own loss continues to decrease throughout, because the frozen quantity is the grafted branch, not the model [1].

That result and this one measure different things — an exact zero under a specific initialization there, a signed decomposition of a nonzero readout here — but both illustrate the same point: a network's output and the gradient signal reaching one of its parts are not the same fact, and neither determines the other.

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